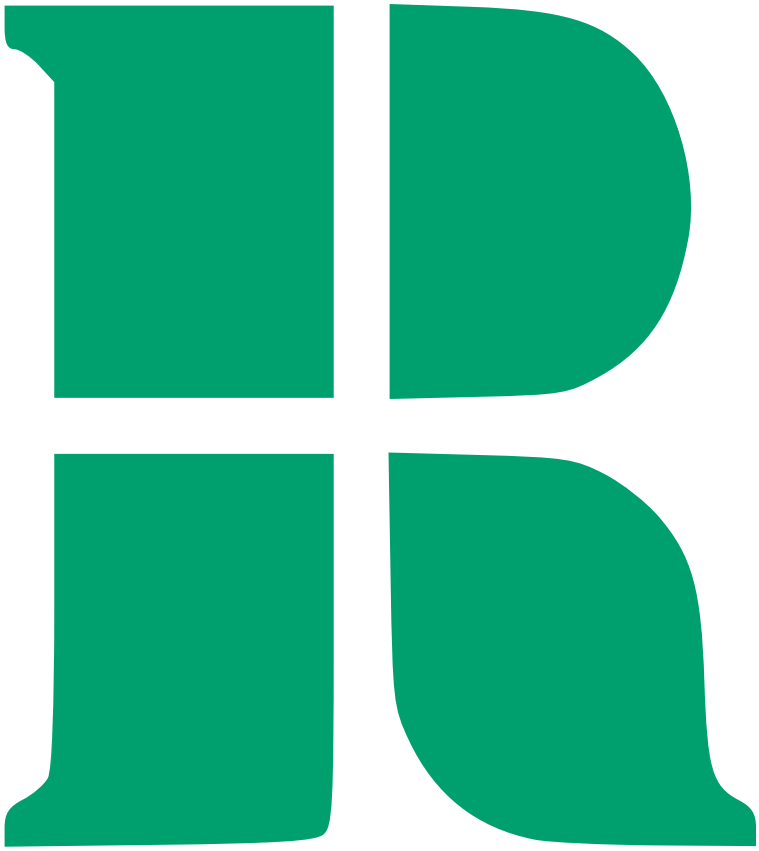


UNIVERSITY OF ROEHAMPTON

# GROUP_COURSE_WORK
# On
# Database Management System

## STUDENTS INVOLVE IN


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## Business Idea ( Bank Management system )

The best method of designing the system is to design the system in a manner that the system can be easily deployed in a bank to manage customer accounts, branches, loans, fixed deposits, and transactional services in an efficient manner. The system has an interface to manage the customer information in an efficient manner, manage loans, fixed deposits and transactions, and conduct core banking operations between branches. Objective:
Its objective is to have safe and secure workers' and clients' working condition and easy banking facility.

Other Features

- Loan & Fixed Deposit Control: Secure handling and processing of the fixed deposits and loans of the customers.
- Transaction & Security Monitoring: Secure monitoring of the cheques, card, and ATMs transactions.
- Employee & Customer Control: Secure control of the workers and customers in rightful manner.
- Branch Operations: Round-the-clock operation of multiple branches with round-the-clock maintenance of availability and continuous accounts.

## Purpose of Work

Systematically apparent feature-based goals of the system are:

1. Customer satisfaction: For ease, simplicity, and security of money process.
2. Appropriate management of credits: For advance status, loan fee, and payment posted for cash handling on time.
3. Online monitoring of exchanges: For online occurrence of the exchange for cheque, card, and ATM.
4. Consolidated Account Management: Customer accounts in office are consolidated together with access control.
5. Branch & Staff Co-ordination: The operations and activities of the office, and staff, are consolidated.
6. Fixed Deposit Accounts: Fixed deposit account, rate of interest, and maturity date are duly maintained.

## Entity-Relationship Diagram (ERD) Analysis

ERD contains 12 entities at major level that are the nature of banking system. ERD is stable and possesses good relation model to conduct banking operations in coordination.

### Entities and Attributes

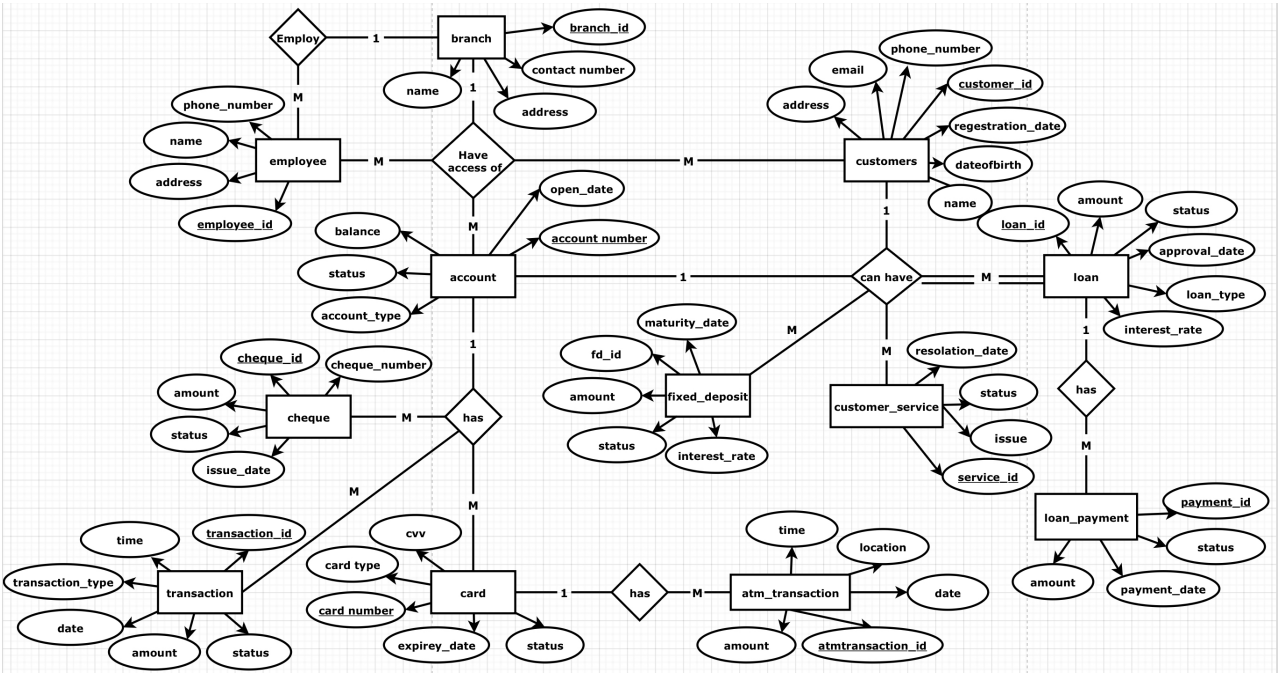
1. Employee (employee_id, name, address, phone_number)
2. Branch (branch_id, name, address, contact_number)
3. Account (account_number, balance, status, account_type, open_date)
4. Cheque (cheque_id, cheque_number, amount, status, issue_date)
5. Transaction (transaction_id, transaction_type, amount, date, time, status)
6. Card (card_number, card_type, expiry_date, CVV, status)

7. ATM Transaction (atmtransaction_id, amount, location, date, time)
8. Customer (customer_id, name, email, phone_number, registration_date, dateofbirth, address)
9. Loan (loan_id, amount, loan_type, approval_date, interest_rate, status)
10. Loan Payment (payment_id, amount, status, payment_date)
11. Fixed Deposit (fd_id, amount, maturity_date, interest_rate, status)
12. Customer Service (service_id, issue, resolution_date, status)

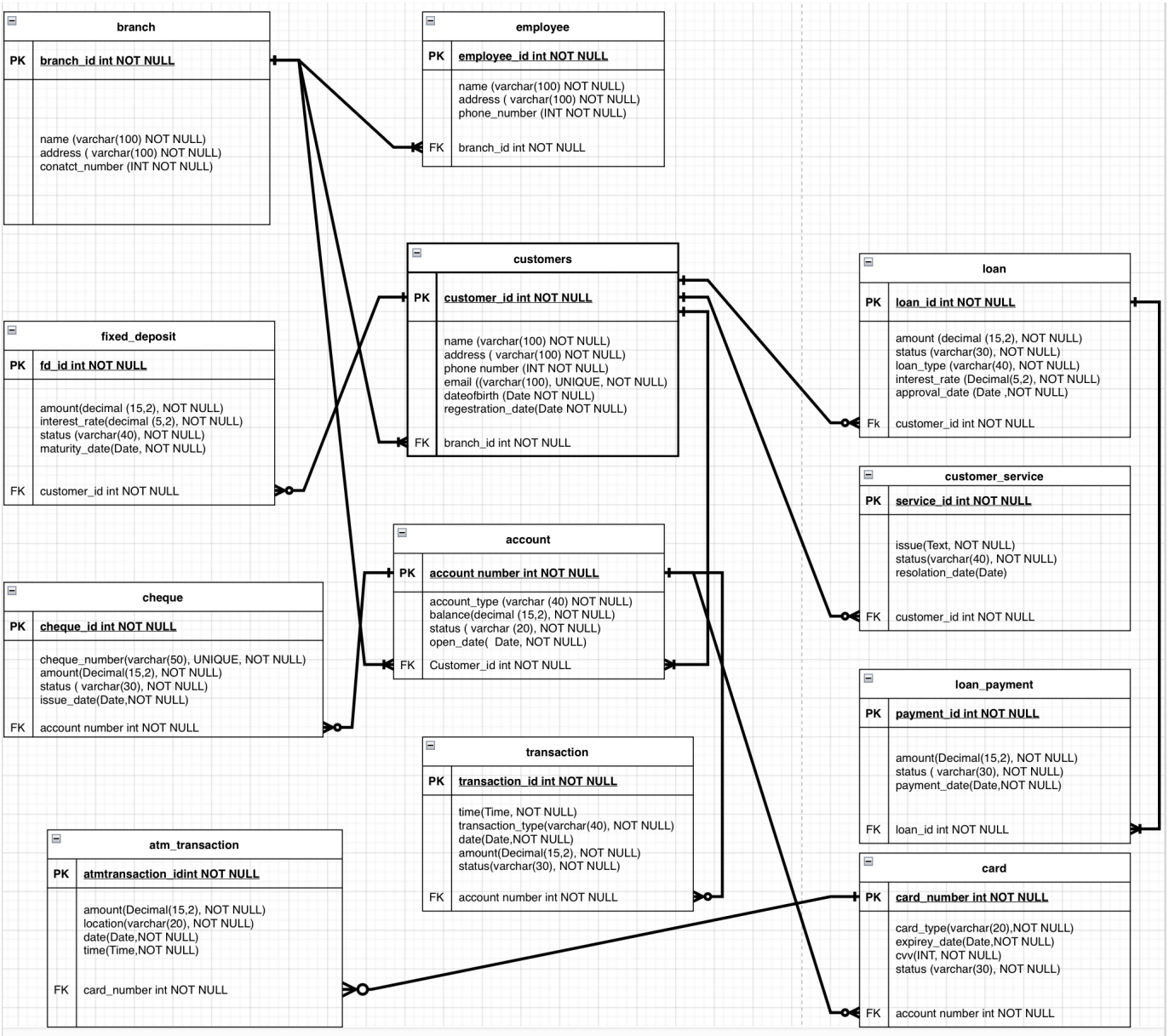
## Relationships and Cardinality

1. Branch & Employee: One employee for a branch (1) but one employee has multiple branches (M).
2. Account & Branch: A branch (M) holds an account, and an account has a branch.
3. Account & Customer: A customer (M) has an account, and an account has a customer.
4. Account & Cheque: A cheque (1) is held in an account (1), and cheques are held in accounts (M).
5. Account & Transaction: Numerous transactions of a single account, a single account for numerous transactions.
6. Card & Account: A single card for a single account, multiple accounts for multiple cards.
7. Card & ATM Transaction: Multiple (M) ATM transactions are kept in a single card (1), and a single (1) card keeps multiple (M) ATM transactions.
8. Customer & Loan: A single customer (1) may have multiple (M) loans, but a single loan (1) is for a single customer (1).
9. Loan & Loan Payment: One payment (1) is for an (M) or many loans, but one payment (1) is for one (1) loan.
10. Customer & Fixed Deposit: Multiple fixed deposits (M) are for one customer (1), and one fixed deposit (1) is for one customer (1).
11. Customer & Customer Service: The majority of service issues (M) in one customer (1), and one service issue (1) is for one customer (1).

## Entity-Relationship Diagram (ERD) Analysis



## Entity-Relationship Physical Diagram (ERD) Analysis




## Conclusion

It is end-to-end bank-class DBMS for fixed deposit, loan, account, and branch-level transaction handling. Fund transfer on a cheque or card basis is a luxury towards ultimate customer delight and customer integrity. ERD modelling is combining the optimal relational model of real money transaction, operations flexibility, and security banking services amongst employees and customers.